

Core PHP

Session 1 – PHP Introduction & Environment Setup

Topics to Cover:

- Introduction to Server-Side Scripting
- History and Features of PHP
- Installing XAMPP
- Understanding htdocs Folder Structure
- Creating First PHP File (`hello.php`)
- Running PHP Files on Localhost

Practical/Demo:

- Run PHP project in browser using localhost
 - Understand browser-server request lifecycle
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Session 2 – PHP Syntax, Variables & Data Types

Topics to Cover:

- PHP Syntax Rules
- Variables and Case Sensitivity
- Data Types:
 - String
 - Integer
 - Float
 - Boolean
 - Array
 - Object
 - Null
- Type Casting
- Dynamic Typing
- `var_dump()` Function

Practical/Demo:

- Display variable types using `var_dump()`

Session 3 – Operators & Expressions

Topics to Cover:

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Concatenation Operators
- Operator Precedence

Practical/Demo:

- Build Bill Calculator Application
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Session 4 – Conditional Statements

Topics to Cover:

- if, else, elseif
- switch Statement
- Nested Conditions
- Ternary Operator (?:)

Practical/Demo:

- Create Student Grading System
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Session 5 – Loops

Topics to Cover:

- while Loop
- do-while Loop
- for Loop
- foreach Loop

- Nested Loops
- break and continue

Practical/Demo:

- Multiplication Tables
 - Pattern Programs
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Session 6 – Functions

Topics to Cover:

- User Defined Functions
- Function Parameters
- Return Values
- Recursive Functions

Practical/Demo:

- Calculator Program
 - Factorial Using Recursion
 - Reverse String Without Function
 - Download File on Button Click
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Session 7 – Arrays

Topics to Cover:

- Indexed Arrays
- Associative Arrays
- Multidimensional Arrays
- Array Traversing
- Array Functions:
 - `array_push()`
 - `sort()`
 - `count()`
 - `in_array()`

Practical/Demo:

- Display Array Values
 - Count Odd & Even Numbers
 - Student Marks Management
 - Shift Zeros to Bottom in Array
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Session 8 – String Functions

Topics to Cover:

- String Concatenation
- strlen()
- substr()
- str_replace()
- explode()
- implode()
- Escaping Characters
- Trimming Whitespace

Practical/Demo:

- Username Formatter Utility
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Session 9 – Working with Forms

Topics to Cover:

- HTML Forms
- GET & POST Methods
- \$_GET and \$_POST
- Input Sanitization

Practical/Demo:

- Contact Us Form
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Session 10 – File Handling

Topics to Cover:

- `fopen()`
- `fwrite()`
- `fread()`
- `fclose()`
- File Upload Using `$_FILES`

Practical/Demo:

- Feedback Saver Application

Session 10 - Live Project

Other Development

- _Resume Building (ATS-friendly)
- LinkedIn Profile Optimization
- GitHub Portfolio Development
- Interview Preparation
- AI Tools for the PHP developers
- Deploy Project on server
- Mock Interviews (Technical + HR)

Duration: 2 Month

Database Management – SQL (Additional Knowledge)

- Session 1: Introduction to Databases Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench
- Session 2: SQL Basics & Commands Trainer Coverage: SQL syntax rules. DDL commands: CREATE, ALTER, DROP. Creating basic users and expenses tables. Demo: Execute commands in MySQL CLI / Workbench
- Session 3: DML Operations Trainer Coverage: Insert, Update, Delete records. WHERE and LIMIT usage. Example: Insert 5 sample users and edit their data
- Session 4: Data Retrieval with SELECT Trainer Coverage: SELECT queries, filtering, sorting, aliases. Aggregate functions (SUM, AVG, COUNT). Demo: Generate “total expenses” by category
- Session 5: Joins and Relationships Trainer Coverage: Foreign keys and primary keys. Inner Join, Left Join, Right Join. Demo: Link users with their transactions table.
- Session 6: Grouping and Subqueries Trainer Coverage: GROUP BY, HAVING. Subqueries and nested queries. Example: “Show users whose average daily spend > ?500”
- Session 7: Constraints & Views Trainer Coverage: Constraints: NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY. Creating and querying VIEWS. Example: Create “ActiveUserView” showing users with >5 transactions